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A IMPLEMENTATION DETAILS

A.1 TASK SPECIFIC ERROR DETECTION

In this work, we specifically address errors produced by modern large language models (LLMs). Given the diverse range of tasks these models are applied to, our focus is on general error detection across all categories, rather than isolating specific types. Prior to the emergence of LLMs, much research targeted error detection for specific tasks, with common examples including grammatical errors (Kasewa et al., 2018; Bell et al., 2019; Cheng & Duan, 2020; Wang & Tan, 2020; Flickinger et al., 2016), spelling mistakes (Mishra & Kaur, 2013), machine translation inaccuracies (Lo, 2019; Pu et al., 2021; Sellam et al., 2020; Gekhman et al., 2020; Rei et al., 2020; 2022a;b), speech recognition faults (Caines et al., 2020; Rao et al., 2020; Li & Wang, 2024; Zhou et al., 2005; Allauzen, 2007; Gekhman et al., 2022; Errattahi et al., 2015; Pellegrini & Trancoso, 2009; Chen et al., 2013), and factual consistency failures (Honovich et al., 2022; Laban et al., 2022; Honovich et al., 2021; Gekhman et al., 2023; Scialom et al., 2021; Kryscinski et al., 2020).

A.2 PROBING: IMPLEMENTATION DETAILS

We examine the intermediate representations of the exact answer tokens generated by a large language model (LLM) during the answer generation process. The intermediate representation selected for this analysis is derived from the output of the final multi-layer perceptron (MLP). This choice is based on preliminary experiments comparing the MLP output, the residual stream, and the attention heads, which showed no significant differences. We leave the in-depth analysis for future work.

For the probing classifier, we employ a logistic regression model from the scikit-learn library (Pedregosa et al., 2011). We used the default hyperparameters, which include a norm penalty of L2 and an LBFGS solver. We initially experimented with other hyper-parameters and did not find a singnificant difference. For each random seed, the dataset was split to training and validation in a 80-20 ratio, and the test dataset was bootstrap sampled.

Obtaining correctness label for the probing dataset. An answer is generally considered correct if it includes the correct answer label and appears before any alternative incorrect labels. We manually analyzed the results of this heuristic to confirm that it is accurate in almost all cases. However, one exception is the Natural Questions with Context (NQ_WC) dataset, where we identified false negatives and thus deployed a more precise validation using an instruct LLM, as demonstrated below:

Evaluate the following answers to questions. For each question you would be given an LLM answer and the correct answer. You would have to determine if the LLM answer is correct or not. If the LLM answer is correct, write '1' and if it is not correct, write '0'. For example:

Question: `[Question 1]`

Ground Truth: `[Gold label 1]`

LLM Answer: `[LLM long answer 1]`

Correctness: `0`

Question: `[Question 2]`

Ground Truth: `[Gold label 2]`

LLM Answer: `[LLM long answer 2]`

Correctness: `1`

Question: `[Question]`

Ground Truth: `[Label]`

LLM Answer: `[LLM long answer]`

Correctness:

Table 3: Success rate of extracting exact answer from a long model answer. Each model is used to extract answers from its own output.

Mistral-7b	Mistral-Instruct-7b	Llama3-8b	Llama3-Instruct-8b
0.99	0.96	0.99	0.95

Detecting and using exact answer tokens. Exact answers are identified from a lengthy generated answer using an external algorithm, which processes the question and the LLM's response, $A(q_i, \hat{y}_i)$, to extract the exact answer. After extraction, we identify the exact answer tokens via a simple search process, focusing on four key tokens: the one before the first exact answer token, the first and last exact answer tokens, and the one after the last.

For the implementation of A that detects the exact locations of answer tokens, we use a combination of heuristic methods and an instruction-tuned LLM. Specifically, when the set of possible answers is finite, we rely on heuristics. For more open-ended scenarios, such as factual questions, we automatically locate the answer if it matches the gold label. Otherwise, we prompt an instruction-tuned LLM, specifically Mistral-7b-Instruct (Jiang et al., 2023), to identify and extract the exact answer substring using the following prompt:

> Extract from the following long answer the short answer, only the relevant tokens. If the long answer does not answer the question, output NO ANSWER.
>
> Q: `[Question 1]`
>
> A: `[LLM long answer 1]`
>
> Exact answer: [Short exact answer 1]
>
> Q: `[Question 2]`
>
> A: `[LLM long answer that does not answer the question]`
>
> Exact answer: NO ANSWER
>
> Q: `[Question]`
>
> A: `[LLM long answer]` Exact answer:

To extract a valid exact answer from a long response, we prompt the instruct LLM up to five times. This process involves verifying that the exact answer is a substring of the long answer unless the instruct LLM indicates that there is no answer. To avoid bias in our probing task, we only retain questions for which a valid exact answer was successfully extracted. This ensures there is no unfair correlation between invalid answers and incorrect answers in the experiments.

We note the following: (a) While it is possible to use an instruct LLM to extract every answer regardless of its correctness, we chose the aforementioned strategy to improve the efficiency of our experiments; (b) This is just one possible implementation. For each LLM, one could use the same LLM to extract its own exact answer token, as demonstrated in a proof-of-concept over 1000 samples of TriviaQA in Table 3. Alternatively, it may be more effective to train a smaller system specifically designed for detecting exact answer tokens, which would be more suitable for real-world scenarios. We choose to keep the extraction process as abstract as possible, as our primary focus is not on the specific implementation, but on analyzing the potential gains from probing these locations.

Additionally, if the exact answer token is not among the first generated tokens, we examine the token immediately preceding it ("before exact answer token"). If the exact answer token is not the last one, we also examine the following token. When the exact answer spans multiple tokens, the first and last exact answer tokens are probed separately.

A.3 DATASETS

We outline here all ten datasets that we investigate in our work. In our analysis, we aimed at covering a wide range of tasks, skills required to solve the tasks, diversity of datasets and as a result also different LLM limitations such as factual inaccuracies (often referred to as "hallucinations"), biases,